

ALGEBRA 1 EXAM

Exercise 1 Give the negation of the following two propositions and indicate true or false.

a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : x + y \leq 0$.

b) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R} : y^2 \geq x$.

c) $\forall y \in \mathbb{R} \setminus \{0\}, \forall x \in \mathbb{R} \setminus \{0\}, \exists z \in \mathbb{R} \setminus \{0\}, z - xy = 0$.

2) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ define by $f(x) = 2x / (1 + x^2)$.

a) f is it injective? surjective?

b) Prove that $f(\mathbb{R}) = [-1, 1]$

c) You can use the restriction $g : [-1, 1] \rightarrow [-1, 1]; g(x) = f(x)$, g is a bijection?

3) In $\mathbb{Z}$, we define : $xRy \Leftrightarrow x - y = 4k$ tel que $k \in \mathbb{Z}$

a) Show that R is an equivalence relation.

b) determine the equivalence classes of 1 and 0.

Exercise 2 Let $a, b \in \mathbb{R}_+$, using absurdity reasoning, show that :

$$\frac{a}{1+b} = \frac{b}{1+a} \Rightarrow a = b$$

Exercise 3 Let E a set and A, B, C three elements of $\mathcal{P}(E)$.

a) Prove that, if $A \cap B = A \cup B$, then $A = B$.

b) Prove that, if $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then $B = C$.